

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SPRING SEMESTER 2026

Assignment (Infinite Series CLO-2) MT-224

1. A ball is dropped from a height of 10 m. Each time it strikes the ground it bounces vertically to a height that is $\frac{3}{4}$ of the preceding height. Find the total distance the ball will travel if it is assumed to bounce infinitely often.
2. A bored student enters the number 0.5 in a calculator display and then repeatedly computes the square of the number in the display.
 - (i) Taking $a_0 = 0.5$, find a formula for the general term of the sequence a_n of numbers that appear in the display.
 - (ii) Find the limit of a_n .
 - (iii) For what values of a_0 will this procedure produce a convergent sequence?
3. An image compression algorithm represents image details with an infinite series of decreasing values. If the first term of the series is 100 and each subsequent term is 75 percent of the previous one, calculate the maximum possible compressed image "information" value, indicating if the algorithm can achieve a finite representation.
4. A software bug is introduced, causing data packets sent by a server to a client to reduce in size by a fixed 10 percent each time they are sent, starting with an initial size of 1MB. If the reduction factor is small enough (e.g., a 10 decrease, meaning 0.9 times the previous size), the total data transmitted will converge to a finite sum, as this is a geometric series with a common ratio $|r| < 1$. However, if the bug causes an increase or a large reduction, the total data transmitted will diverge to infinity, as the sum would be infinite. Computer engineers must analyze the series to determine if the data loss is within acceptable limits (convergent) or will continue to grow indefinitely (divergent), impacting system stability.
5. Consider an electrical network with an impedance function $Z(s) = \frac{s+2}{(s+1)(s+3)}$. Determine the poles and zeroes of the impedance function.

A digital filter is described by the transfer function $H(z) = \frac{z-1}{z^2-0.5z-0.6}$. Identify the poles and zeroes of this digital filter.
6. System Stability: A digital filter's transfer function is given by $H(z) = \frac{z^2-5z+6}{z^2-3z+2}$. Identify the poles and zeros of this system. Determine if the system is stable by examining the location of the poles. Explain your reasoning in terms of the system's stability conditions. Concept: Z-transform, stability, poles in the complex plane.